

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T50 · 3D Pythagoras and Trigonometry

Questions

Student book · questions with writing space

■ 11 pages

FAMILY MAP

3D Pythagoras Trigonometry

F01

Resolve multi-stage solid geometry

5 Q | 23 marks

F02

Use trigonometry in three dimensions

3 Q | 12 marks

Working Unit 26

4MA1-2H Q24

UNIT 1
4 marks

24 The diagram shows a square-based pyramid $ABCDE$

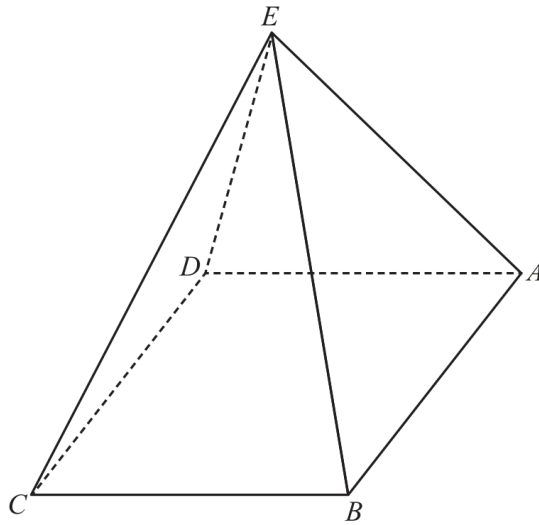


Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

M is the centre of the horizontal square base $ABCD$

Q is the midpoint of AB

Angle $EQM = 80^\circ$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

Question 17

4MA1-1H Q21

MODULAR UNIT 1

5 marks

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

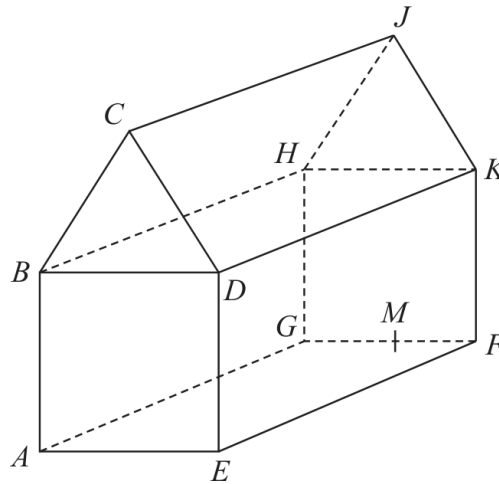


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

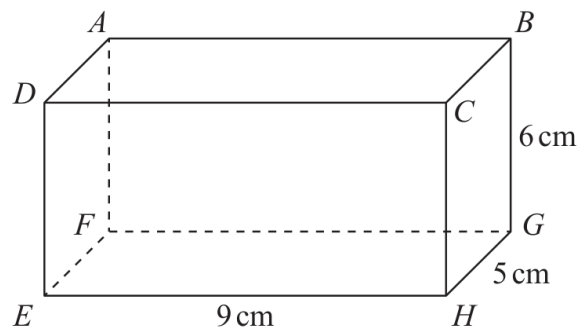
M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

Diagram **NOT**
accurately drawn



Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \qquad AB = 10 \text{ cm} \qquad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

18 The diagram shows cuboid $ABCDEFGH$.

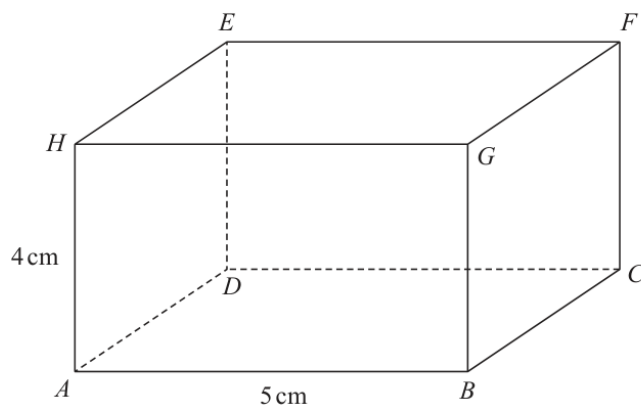


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

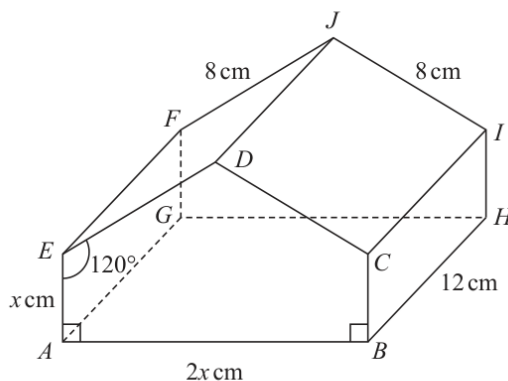
The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

Diagram **NOT**
accurately drawn



Give your answer correct to 3 significant figures.

(Total for Question 23 is 5 marks)

$$AB = 10 \text{ cm} \qquad BG = 14 \text{ cm}$$
$$BC = \frac{1}{5}AD$$

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm³

(Total for Question 26 is 5 marks)